DIGITAL FORENSIC EXAMINATION REPORT

Analysis of a USB Flash Drive Image

Case Reference: USB Flash Drive Forensic Examination
Evidence Item: flash.dd (Raw Forensic Image)
Examination Platform: SANS Investigative Forensic Toolkit (SIFT) Workstation
Tool Suite: The Sleuth Kit (TSK)

1. Scenario

A USB flash drive was acquired and preserved as a forensic image named flash.dd for digital forensic examination. The objective was to analyze the image without altering the original evidence, identify the partition layout and file system, examine stored files, recover selected evidence, and identify deleted files using The Sleuth Kit (TSK) on a SIFT Workstation.

2. Methodology

The examination followed standard digital forensic procedures using The Sleuth Kit (TSK). The investigation included image verification, partition identification, file system identification, file listing, file recovery, and deleted file analysis. All analysis was performed on the forensic image to preserve the integrity of the original USB device.

3. Task 1 – Verify the Image

Objective

Verify the Image.

Command Used

```
img_stat flash.dd
```

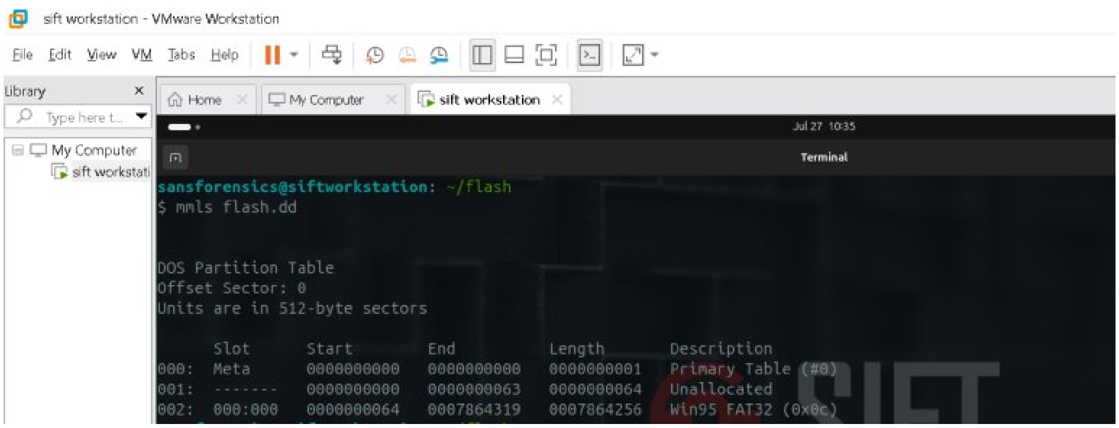
Findings

- Image Type: Raw

- Image Size: 4,026,531,840 bytes
- Sector Size: 512 bytes

Analysis

The output confirms that flash.dd is a valid raw forensic image suitable for forensic examination.
Figure 1: Insert screenshot/evidence here.



4. Task 2 – Locate the Partition

Objective

Locate the Partition.

Command Used

mmls flash.dd

Findings

- Partition Table Type: DOS (MBR)
- Start Sector: 64

Analysis

The partition begins at sector 64, which was used as the offset for subsequent commands.
Figure 2: Insert screenshot/evidence here.

5. Task 3 – Identify the File System

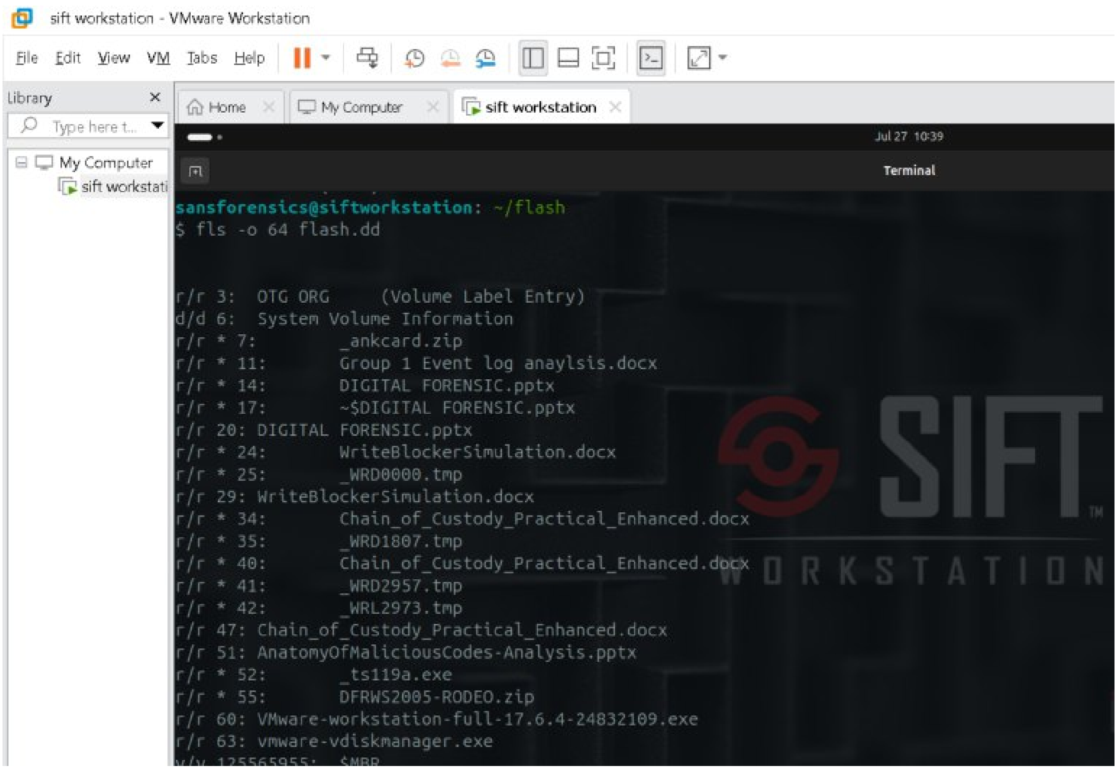
Objective
Identify the File System.

Command Used

fsstat -o 64 flash.dd

Findings
- File System: FAT32
- Volume Label: OTG ORG
- Sector Size: 512 bytes

Analysis
The image contains a FAT32 file system with the volume label OTG ORG.
Figure 3: Insert screenshot/evidence here.



6. Task 4 – List Files

Objective

List Files.

Command Used

fls -o 64 flash.dd

Findings

- DIGITAL FORENSIC.pptx
- Group 1 Event log analysis.docx
- WriteBlockerSimulation.docx
- Chain_of_Custody_Practical_Enhanced.docx
- AnatomyOfMaliciousCodes-Analysis.pptx
- VMware-workstation-full-17.6.4-24832109.exe
- VMware-vdiskmanager.exe

- ts119a.exe
- DFRWS2005-RODEO.zip

Analysis
The file listing shows documents, presentations, executables, and archive files stored on the USB image.
Figure 4: Insert screenshot/evidence here.

7. Task 5 – Recover a File

Objective
Recover a File.

Command Used

```
icat -o 64 flash.dd 52 > ts119a.exe
```

Findings
- Recovered file: ts119a.exe

Analysis
The executable was successfully recovered from the forensic image without modifying the original evidence.
Figure 5: Insert screenshot/evidence here.

8. Task 6 – Identify Deleted Files

Objective
Identify Deleted Files.

Command Used

fls -r -d -o 64 flash.dd

Findings

- _ankcard.zip
- Group 1 Event log analysis.docx
- DIGITAL FORENSIC.pptx
- ~$DIGITAL FORENSIC.pptx
- WriteBlockerSimulation.docx
- Chain_of_Custody_Practical_Enhanced.docx
- ts119a.exe
- DFRWS2005-RODEO.zip

Analysis

Several deleted files remain identifiable within the forensic image and may be recoverable.

Figure 6: Insert screenshot/evidence here.

Investigation Questions

Question	Answer
What image format was examined?	Raw (.dd) image named flash.dd
Which partition table was identified?	DOS (MBR)
What is the partition start sector?	64
Which file system was detected?	FAT32
What is the volume label?	OTG ORG

Which command lists files?	fls -o 64 flash.dd
Which command recovers a file?	icat -o 64 flash.dd 52 > ts119a.exe
Which command identifies deleted files?	fls -r -d -o 64 flash.dd

Conclusion

The forensic examination of flash.dd verified a valid raw forensic image, identified a DOS (MBR) partition with a start sector of 64, and confirmed a FAT32 file system labeled OTG ORG. Multiple files were identified, ts119a.exe was successfully recovered, and deleted file entries were identified. The examination demonstrates the effective use of The Sleuth Kit for forensic analysis while preserving evidence integrity.